ROB5014

Mathematics and Simulation for Robotics

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Agenda

- Numerical Integration
- Previously, we discussed a dynamics formulation to PSG
- However, we still use simple Euler integration
  - P(k+1)=P(k)+h*V(k)
- Is this the only way?

Motivation

- Virtual world, Digital twin
  - dynamic models
    - $m\ddot{x} + b\dot{x} + kx = f + f_c$
    - $J\dot{\omega} + \omega \times J\omega + b\omega + k\omega = \tau + \tau_c$
    - $\dot{R} = R\Omega$
  - real-time update
- The key is the time integration
  - Euler integrator
    - $v_i = \frac{x_{i+1} - x_i}{T} \to x_{i+1} = x_i + Tv_i$
- Generalization of numerical integration
  - $\frac{dx}{dt} = f(t, x) \Rightarrow x_{i+1} = x_i + f(t_i, x_i)h$ where $h = t_{i+1} - t_i$

Taylor Series and Runge–Kutta 2nd order Method

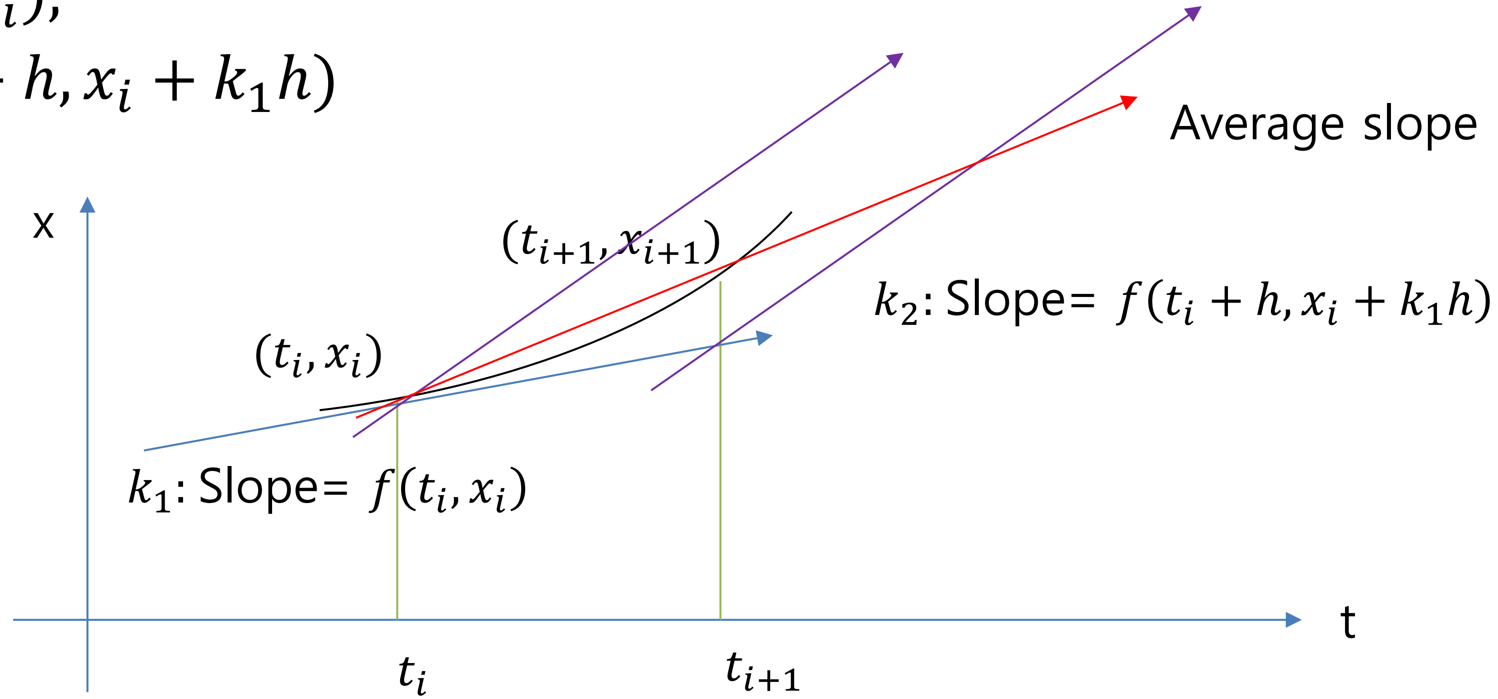
- Taylor series
  - $x_{i+1} = x_i + \frac{dx}{dt}(t_i, x_i)(t_{i+1} - t_i) + \frac{1}{2!}\frac{d^2x}{dt^2}(t_i, x_i)(t_{i+1} - t_i)^2 + O(3)$
  - Recall that $\frac{dx}{dt} = f(t, x)$
  - Thus,
  - $x_{i+1} = x_i + f(t_i, x_i)h$: Euler's method
- Runge–Kutta 2nd Method
  - Second order Taylor series approximation for integration
  - The question is how to obtain the second order derivatives

Runge–Kutta 2nd Method

- $x_{i+1} = x_i + (a_1 k_1 + a_2 k_2)h$ (What we know is that $\frac{dx}{dt} = f(t, x)$)
- Where
  - $k_1 = f(t_i, x_i)$
  - $k_2 = f(t_i + p_1 h, x_i + q_{11} k_1 h)$
  - 4 Unknows: a_1, a_2, p_1, q_{11}
  - But, we have only 3 equations.
  - Depending on the choice of a_2, we have different names of methods:
    - Heun's ($a_2 = \frac{1}{2}$)
    - Midpoint ($a_2 = 1$)
    - Ralston's ($a_2 = \frac{2}{3}$)

Runge–Kutta 2nd Method

- Heun's method: $a_2 = \frac{1}{2}$, $a_1 = \frac{1}{2}$, $p_1 = 1$, $q_{11} = 1$
  - $x_{i+1} = x_i + \left(\frac{1}{2}k_1 + \frac{1}{2}k_2\right)h$
  - $k_1 = f(t_i, x_i)$,
  - $k_2 = f(t_i + h, x_i + k_1 h)$



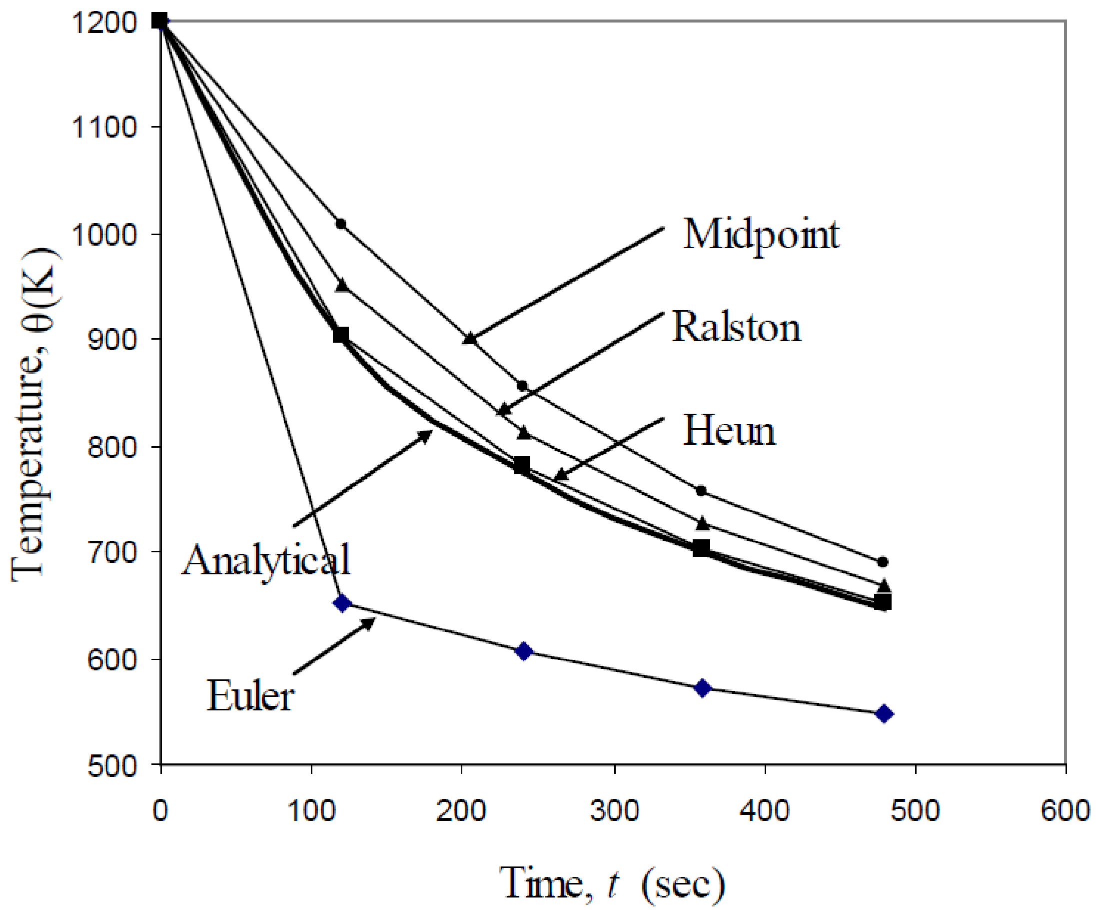
Variations

- Midpoint method: $a_2 = 1, a_1 = 0, p_1 = \frac{1}{2}, q_{11} = \frac{1}{2}$
  - $x_{i+1} = x_i + k_2 h$
  - $k_1 = f(t_i, x_i)$
  - $k_2 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_1 h)$
- Ralston's method: $a_2 = \frac{2}{3}, a_1 = \frac{1}{3}, p_1 = \frac{3}{4}, q_{11} = \frac{3}{4}$
  - $x_{i+1} = x_i + \left(\frac{1}{3}k_1 + \frac{2}{3}k_2\right)h$
  - $k_1 = f(t_i, x_i)$
  - $k_2 = f(t_i + \frac{3}{4}h, x_i + \frac{3}{4}k_1 h)$

Example

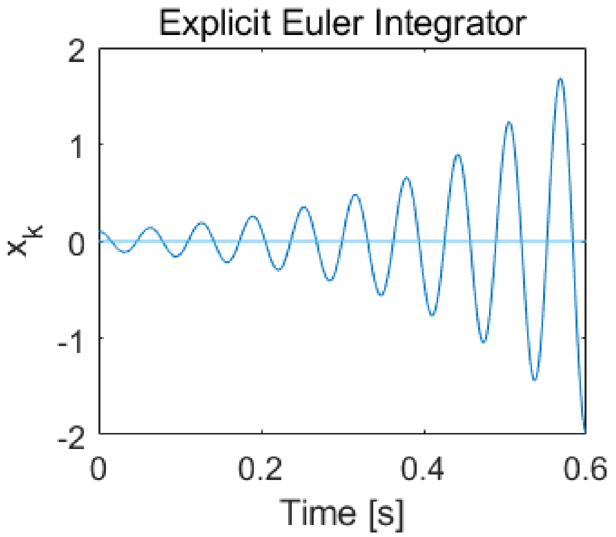
- $\frac{d\theta}{dt} = -2.2067 \times 10^{-12}(\theta^4 - 81 \times 10^8)$
- To be accurate, we need a small h
- Which requires more time to compute

Step size, h	$\theta(480)$			
	Euler	Heun	Midpoint	Ralston
480	-987.84	-393.87	1208.4	449.78
240	110.32	584.27	976.87	690.01
120	546.77	651.35	690.20	667.71
60	614.97	649.91	654.85	652.25
30	632.77	648.21	649.02	648.61



Problem and Question that leads to a Solution

- Problem of Euler Integrator
  - When a mass–spring–damper problem is stiff (m=0.01kg, k=100N/m)
  - Mass is too small
  - Spring stiffness is very high
  - Causing numerical integration inaccuracy
- What is the condition that guarantees the passivity of the integration such that we have $L: (x_i, v_i, f_i) \rightarrow (x_{i+1}, v_{i+1})$?



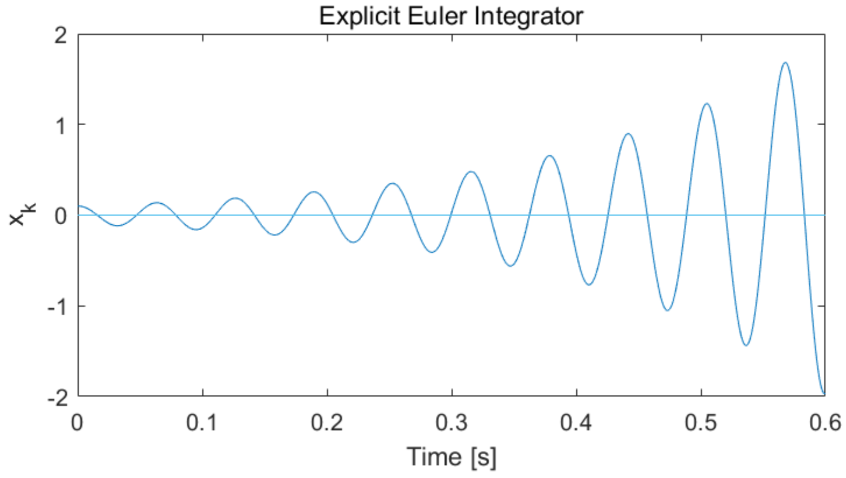
Passivity: $\int_0^T u^T y dt \geq E(T) - E(0), \forall T \geq 0$

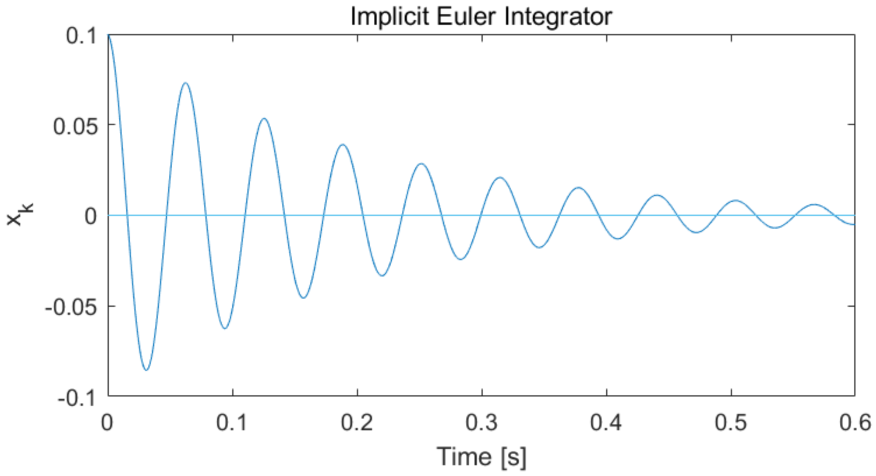
Integrators

- Continue the mass–spring–damper example
- Explicit Euler Integrator
  - $m\frac{v_{i+1}-v_i}{T} + kx_i = f_i,\ x_{i+1} = x_i + Tv_i$
- Symplectic Euler Integrator
  - $m\frac{v_{i+1}-v_i}{T} + kx_i = f_i,\ x_{i+1} = x_i + Tv_{i+1}$
- Implicit Euler Integrator
  - $m\frac{v_{i+1}-v_i}{T} + kx_{i+1} = f_i,\ x_{i+1} = x_i + Tv_{i+1}$
- Passive Midpoint Integrator
  - $m\frac{v_{i+1}-v_i}{T} + k\hat{x}_i = f_i,\ x_{i+1} = x_i + T\hat{v}_i,\ \hat{x}_i = \frac{x_{i+1}+x_i}{2},\ \hat{v}_i = \frac{v_{i+1}+v_i}{2}$

Simulation

- Simple harmonic motion
  - mass m = 0.01kg
  - spring constant k = 100N/m
  - No external force f = 0;
  - Initial condition $x_0 = 0.1m$
- Methods tend to fail.
  - One remedy may reduce the time scale $(T = 0.0001)$
  - However, this is NOT desired due to the increase of the computational cost
  - Better idea is to make the integrator stable (passive)





Proof of Passivity (HW)

- We begin with the dynamics and energy computation
  - $m\dot{v} + bv + kx = f$
- Prove that
  - $\int_{t_0}^{t_f} fv dt = \sum_{i=0}^{N-1} f_i \hat{v}_i T_i = \frac{1}{2}m(v_N^2 - v_0^2) + \frac{1}{2}k(x_N^2 - x_0^2) + \sum_{i=0}^{N-1} b\hat{v}_i^2 T_i \geq -\frac{1}{2}mv_0^2 - \frac{1}{2}kx_0^2 = -E_0$
  - where $\hat{v}_i \coloneqq \frac{v_{i+1}+v_i}{2}$, $\hat{x}_i = \frac{x_{i+1}+x_i}{2}$

Passivity Properties

- EEI
  - $\sum_{i=0}^{N-1} f_i v_i T_i \leq \frac{1}{2} m(v_N^2 - v_0^2) + \frac{1}{2} k(x_N^2 - x_0^2)$ which does not guarantee the passivity
- IEI
  - $\sum_{i=0}^{N-1} f_i v_i T_i \geq \frac{1}{2} m(v_N^2 - v_0^2) + \frac{1}{2} k(x_N^2 - x_0^2)$ which is passive, but dissipative (loosing energy), thus cannot simulate the harmonic behavior
- SEI
  - in definite, meaning not passive nor non-passive.
  - Conditionally stable depending on the parameters
- PMI
  - $\sum_{i=0}^{N-1} f_i \hat{v}_i T_i = \frac{1}{2} m(v_N^2 - v_0^2) + \frac{1}{2} k(x_N^2 - x_0^2)$ passive and lossless.

Thank you!